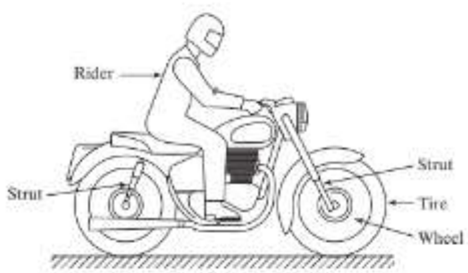
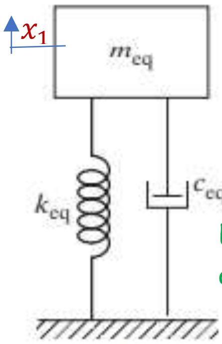
Mathematical Modelling

Let us consider a case of testing the rider comfort of a two wheeler. When the vehicle moves over rough road or hits a road bump, the system (vehicle+rider) vibrates in the vertical direction. We need to evaluate the amplitudes of displacement, velocity, and acceleration of the system in the vertical direction to decide whether any re-designing of the suspension system is required or not.

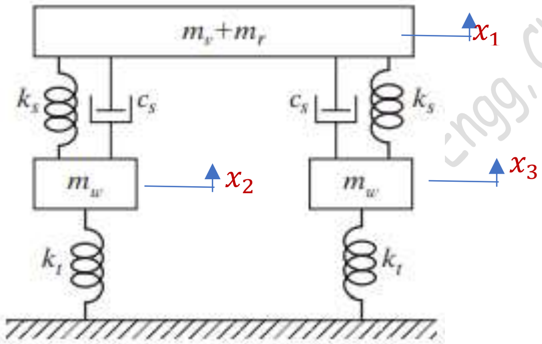


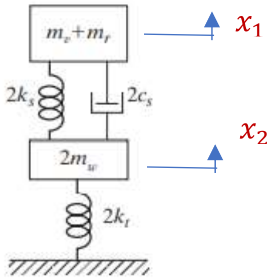


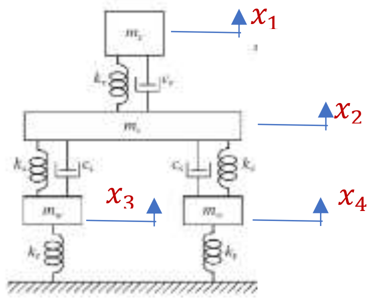
Equivalent mass of rider, vehicle, and wheels

Equivalent stiffness of seat, suspension springs, and tyres

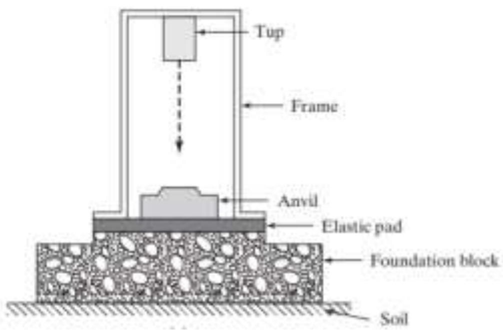
Equivalent damping of seat and suspension

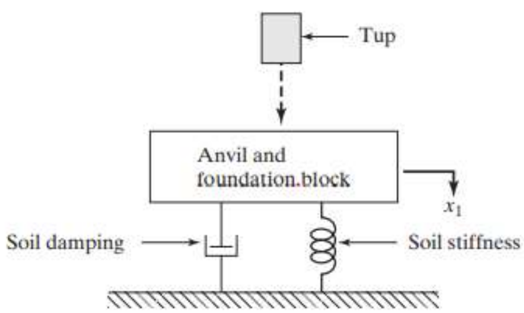


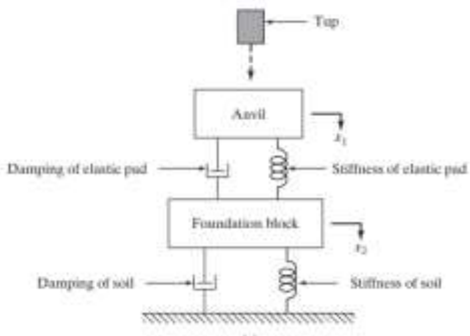




Consider another example of a forging press. The forging hmmer (tup) falls through height on to anvil and because of the impact the system vibrates. The modelling of the system is shown in the subsequent figures.







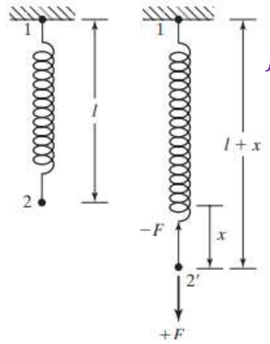
Spring Elements

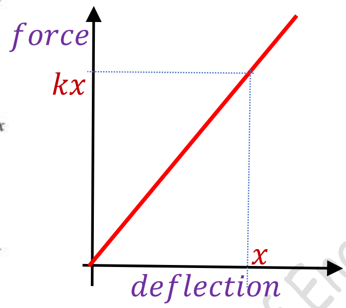
In the context of vibration, any component that stores strain energy/potential energy is considered as a spring element. It includes not only helical springs or leaf springs but bars, shafts, columns, beams, cables, plates etc. The spring element is represented by its stiffness (spring constant). It is defined as the force required to produce unit deflection of the spring.

If the spring element is linear, the spring force $F = kx$.

The energy stored in the spring element= work done on the spring=area of the force-displacement diagram.

$$U = \frac{1}{2}kx^2$$





Axial spring stiffness of a bar

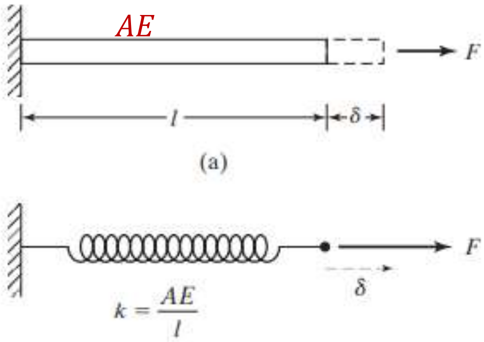
Consider a bar of length l, cross sectional area A, and modulus of elesticity E is sublected to an axial force F.

From solid mechanics, we know that the elongation

$$\delta = \frac{Fl}{AE}$$

Force F produces a deflection δ. Therefore, the axial stiffness of the bar is given by

$$k = \frac{F}{\delta} = \frac{AE}{l}$$

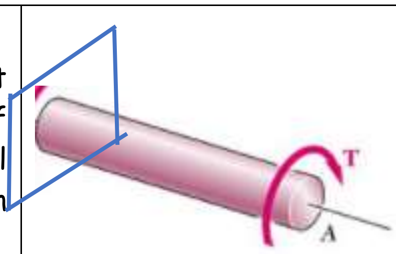


Torsional stiffness of circular shafts

Consider that a torque T is applied to a circular shaft of length l, modulus of rigidity G, and polar moment of inertia of cross section with respect to its geometrical axis (torque axis) J. Let θ be the angular deflection produced. We have the torsion formula

$$\frac{T}{J} = \frac{G\theta}{l} \quad \ggg\gg \quad \frac{T}{\theta} = \frac{GJ}{l}$$

The torsional stiffness, $k = \frac{GJ}{l}$

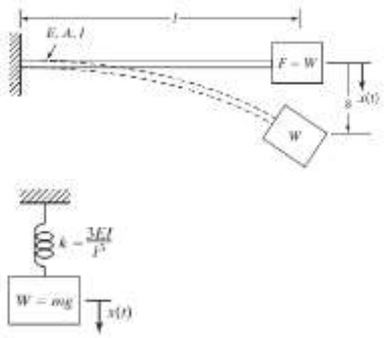


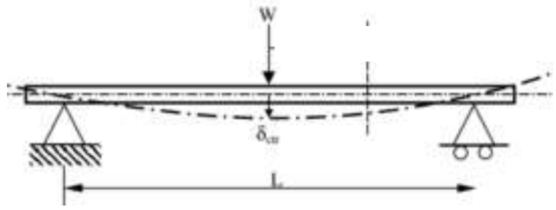
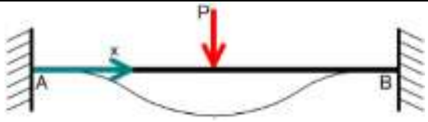
Flexural stiffness of a cantilever beam

Consider a cantilever beam of length l, moment of inertia of the cross section about the neutral axis I, modulus of elasticity E subjected to concentrated load at the free end.

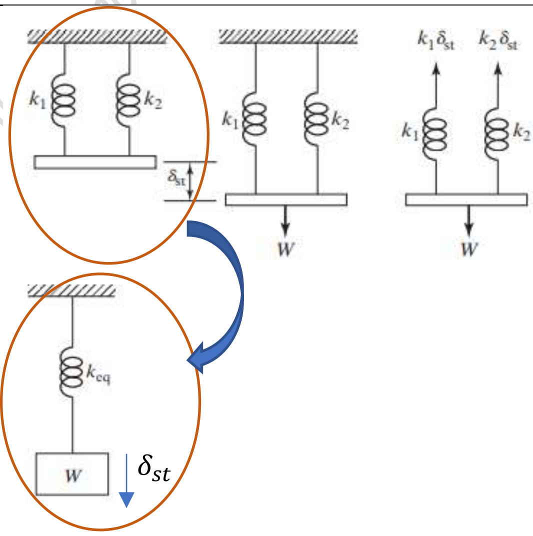
$$\delta = \frac{Wl^3}{3EI} \gg\gg \frac{W}{\delta} = \frac{3EI}{l^3}$$

Bending stiffness of a cantilever beam with point load at its end, $k = \frac{3EI}{l^3}$



Flexural stiffness of a simply supported beam with central point load $$\delta = \frac{Wl^3}{48EI} \ggg \frac{W}{\delta} = \frac{48EI}{l^3}$$ Bending stiffness of a simply supported beam with central load, $k = \frac{48EI}{l^3}$	
Flexural stiffness of a fixed-fixed beam with central point load $$\delta = \frac{Wl^3}{192EI} \ggg \frac{W}{\delta} = \frac{192EI}{l^3}$$ Bending stiffness of a fixed-fixed beam with central load, $k = \frac{192EI}{l^3}$	

Equivalent spring stiffness

Spring elements in parallel Consider two springs of stiffnesses k_1 and $k_2$connected in parallel and subjected to a force W. Both the springs undergo same deflection δ_{st}. But, the force on each spring will be different. Now, let us replace these two springs by an equivalent spring having the same deflection δ_{st}. If both springs are to equivalent, the strain energy of the two systems should be the same. Strain energy of the parallel springs, $U = U_1 + U_2$ $$U = \frac{1}{2}\ k_1\delta_{st}{}^2 + \frac{1}{2}\ k_2\delta_{st}{}^2$$	
Strain energy of the equivalent system $$U_{eq} = \frac{1}{2}\ k_{eq}\delta_{st}{}^2$$ If both are to be equivalent, $U_{eq} = U$ $$\frac{1}{2}\ k_{eq}\delta_{st}{}^2 = \frac{1}{2}\ k_1\delta_{st}{}^2 + \frac{1}{2}\ k_2\delta_{st}{}^2 \quad \ggg\ggg \ k_{eq} = k_1 + k_2$$	

Vibration & Noise control.....Dr. Biju N., Professor, School of Engg, CUSAT

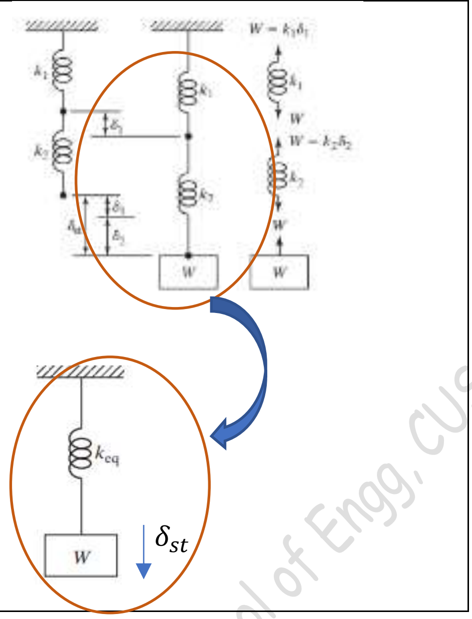
Spring elements in series

Consider two springs of stiffnesses k_1 and k_2 connected in series and subjected to a force W. Each spring will have a deflection of δ_1 and δ_2 so that $\delta_{st} = \delta_1 + \delta_2$. Here, we can see that the forces on each spring will be the same but the deflections are different.

Now, let us replace these two springs by an equivalent spring having the same deflection δ_{st}. If both springs are to equivalent, the strain energy of the two systems should be the same.

Strain energy of the parallel springs, $U = U_1 + U_2$

$$U = \frac{1}{2}\, k_1 {\delta_1}^2 + \frac{1}{2}\, k_2 {\delta_2}^2 = \frac{1}{2}\, k_1 \left(\frac{W}{k_1}\right)^2 + \frac{1}{2}\, k_2 \left(\frac{W}{k_2}\right)^2$$



Strain energy of the equivalent system

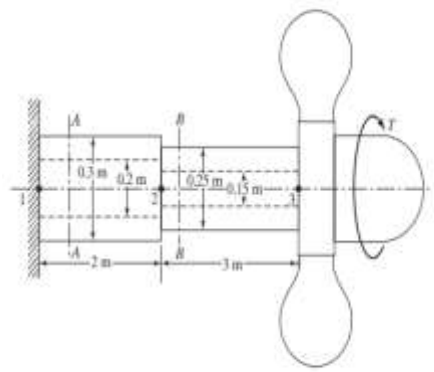
$$U_{eq} = \frac{1}{2}\, k_{eq} {\delta_{st}}^2 = \frac{1}{2}\, k_{eq} \left(\frac{W}{k_{eq}}\right)^2$$

If both are to be equivalent, $U_{eq} = U$

$$\frac{1}{2}\, k_{eq} \left(\frac{W}{k_{eq}}\right)^2 = \frac{1}{2}\, k_1 \left(\frac{W}{k_1}\right)^2 + \frac{1}{2}\, k_2 \left(\frac{W}{k_2}\right)^2 \ggg\ggg \frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

Example 1

Determine the equivalent stiffness of the propellor shaft shown in figure. The modulus rigidity of the shaft material is 80 GPa.



Solution:

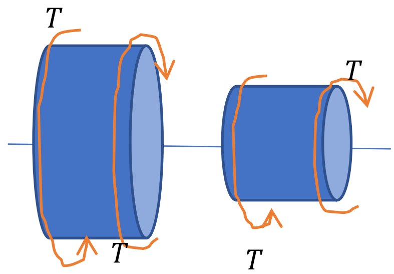
When we draw the free body diagram of the two stepped propellor shaft, it is clear that the torque acting on each step is the same. But, the angular deformations will be different. Hence, the two spring elements can be considered as in series.

Since, the shafts are subjected to torque T, the torsional stiffness of bigger shaft

$$k_1 = \frac{GJ_1}{l_1} = \frac{80 \times 10^9 \times \frac{\pi}{32}(0.3^4 - 0.2^4)}{2} = 2.55 \times 10^7 Nm/rad$$

Similarly,

$$k_2 = \frac{GJ_2}{l_2} = \frac{80 \times 10^9 \times \frac{\pi}{32}(0.25^4 - 0.15^4)}{3}$$
$$= 0.89 \times 10^7 Nm/rad$$



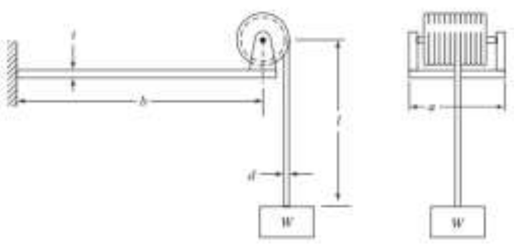
Since the spring elements are in series, $\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$

$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2} = \frac{2.55 \times 10^7 \times 0.89 \times 10^7}{2.55 \times 10^7 + 0.89 \times 10^7} = 0.659 \times 10^7\ N.m/rad$$

Vibration & Noise Control.........Dr. Biju N., Professor, School of Engg, CUSAT

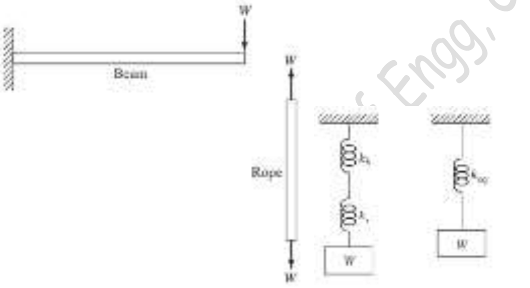
Example 2

A hoisting drum, carrying a steel wire rope, is mounted at the end of a cantilever beam as shown in figure. Determine the equivalent spring constant of the system when the suspended length of the wire rope is l. Assusme that the net cross-sectional diameter of the wire rope is d and the Young's modulus of the beam and the wire rope is E



$d = 2\ cm, l = 3m, b = 1m, t = 1cm,$
$a = 5\ cm, E = 200\ GPa$

If we look at the free body diagram, we can see that the force acting on the wire is W, and the force acting on the end of the cantilever is also W. That is, the forces acting on both the elements are the same and hence they are in series.



Axial stiffness of the bar,

$$k_1 = \frac{AE}{l} = \frac{\pi d^2 E}{4l} = \frac{\pi \times 0.02^2 \times 200 \times 10^9}{4 \times 3} = 20.9439 \times 10^6\ N/m$$

Flexural stiffness of the cantilever beam,

$$k_2 = \frac{3EI}{b^3} = \frac{3E}{b^3}\left(\frac{at^3}{12}\right) = \frac{3 \times 200 \times 10^9}{1^3}\left(\frac{0.05 \times 0.01^3}{12}\right) = 0.0025 \times 10^6\ N/m$$

Since they are in series, the equialent stiffness,

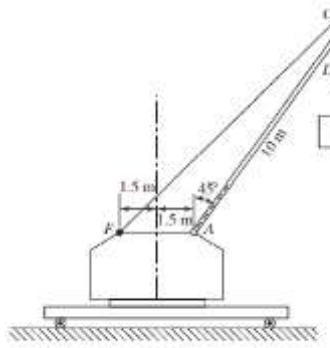
$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2} = \frac{20.9439 \times 10^6 \times 0.0025 \times 10^6}{20.9439 \times 10^6 + 0.0025 \times 10^6} = 0.002499 \times 10^6\ N/m$$

Vibration & Noise Control.......Dr. Biju N., Professor, School of Engg, CUSAT

Example 3

The boom AB of the crane shown is a uniform steel bar of length $10\ m$ and area of cross section $2500\ m^2$. A weight W is suspended while ethe crane is stationary. The cable $CDEBF$ is made of steel and has a cross sectional area of $100\ mm^2$. Neglecting the effect of the cable $CDEB$, find the equivalent stiffness of the system in the vertical direction. Assume the modulus of elasticity of steel, $E = 200\ GPa$.

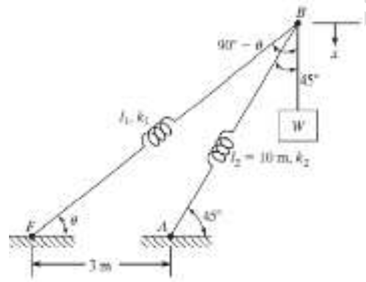


Consider the simplified diagram of the crane. The cable and the boom are shown as two spring elements of stiffnesses k_1 and k_2 respectively. Let the end B has a vertical displacemet x. The deformation of the cable (spring k_1) is,

$$x_1 = x\cos(90-\theta) = x\sin\theta$$

The deformation of the boom (spring k_2),

$$x_2 = x\cos 45$$

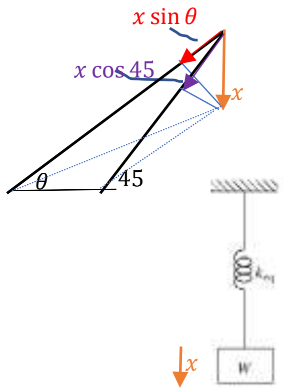


The forces acting on the boom and the cable are along their axes. Therefore, the axial stiffness of the cable, $k_1 = \frac{A_1E}{l_1}$ and the axial stiffness of the boom $k_2 = \frac{A_2E}{l_2}$. Here, both the forces and displacements are different on the two spring elements, we can not consider them as either parallel or series.

$$l_1{}^2 = 3^2 + 10^2 - 2\times 3\times 10\times\cos 135 \ggg l_1 = 12.306\ m$$

We can also write,

$$10^2 = 3^2 + l_1{}^2 - 2\times 3\times l_1\times\cos\theta \ggg \theta = 35.08°$$



$$k_1 = \frac{A_1E}{l_1} = \frac{100\times10^{-6}\times 200\times10^{9}}{12.306} = 1.625\times10^6\ \frac{N}{m} \quad \lll\ggg \quad x_1 = x\sin\theta = 0.5747x$$

$$k_2 = \frac{A_2E}{l_2} = \frac{2500\times10^{-6}\times 200\times10^{9}}{10} = 50\times10^6\ \frac{N}{m} \quad \lll\ggg \quad x_2 = x\cos 45 = 0.7071x$$

The strain energy stored in spring element 1,

$$U_1 = \frac{1}{2}k_1x_1{}^2 = \frac{1}{2}\times 1.625\times10^6\times(0.5747x)^2 = 0.2684x^2\times10^6$$

The strain energy stored in spring element 2,

$$U_2 = \frac{1}{2}k_2x_2{}^2 = \frac{1}{2}\times 50\times10^6\times(0.7071x)^2 = 12.4998x^2\times10^6$$

The total strain energy of the system, $U = U_1 + U_2 = 12.7646x^2\times10^6$

If the two springs are replaced by a single equivalent spring of stiffness k_{eq} as shown, the strain energy of the equicalent system, $U_{eq} = \frac{1}{2}k_{eq}x^2$. Since both the systems are equivalent, $U_{eq} = U$.

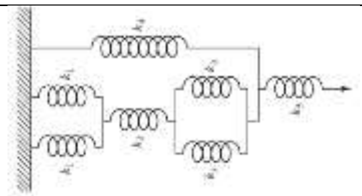
$$\frac{1}{2}k_{eq}x^2 = 12.7646x^2\times10^6$$

$$k_{eq} = 25.5291\times10^6\ N/m$$

Vibration & Noise Control.......Dr. Biju N., Professor, CUSAT

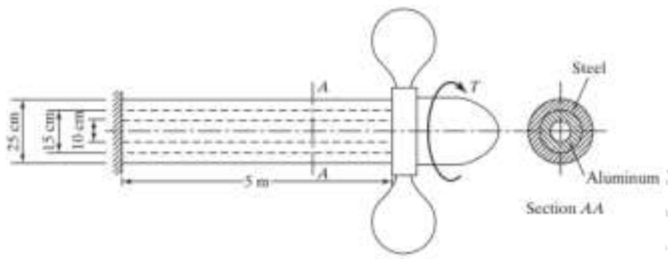
#Exercise 1

Calculate the equivalent spring constant of the system shown assuming all the springs are having the same spring constant k.



#Exercise 2

Calculate the equivalent torsional spring constant of the composite propellor shaft. Take the modulus of rigidity of steel as $75\ GPa$ and that of aluminium as $25\ GPa$



Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition

Vibration & Noise Control..........Dr. Biju N, Professor, School of Engg., CUSAT

Vibration & Noise Control..........Dr. Biju N, Professor, School of Engg, CUSAT